

RUDRA PRAJAPATI

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LinkedIn

GitHub

CodeForces

Education

Sardar Vallabhbhai National Institute of Technology, Surat

2024 - 2028

B.Tech in Electronics and Communication Engineering

8.74/10 CGPA

Crystal School

2023 - 2024

Class XII (Gujarat State Board)

80.15/100 %

Crystal School

2021 - 2022

Class X (Gujarat State Board)

95/100 %

Projects

Solvexe

Live Source Code

- Built Algorithmic Arena, a programming platform with **Codebox**-powered code execution and automated **boilerplate code generation** across multiple languages.
- Designed an **event-driven submission pipeline** using **RabbitMQ** and **Redis**, processing **3 submissions concurrently** per worker while keeping API response times under **350ms**.
- Built a plagiarism detection system that compares code submissions for similarity, automatically flagging matches above **85%** to catch duplicate solutions.
- Built a Codeforces-style **ELO rating system** and real-time leaderboard using **Redis Sorted Sets**, with a rate limiter capping submissions at **5 requests per 10 seconds**.

AI MockInterview

Source Code

- Built **Mock AI Interview**, a self-hostable real-time voice AI platform using **WebRTC** and **FastAPI**, achieving **sub-2.5s end-to-end latency** (STT→LLM→TTS).
- Built a real-time speech-to-speech AI pipeline using **Whisper**, **Ollama**, and **Piper TTS**, streaming responses sentence-by-sentence to reduce latency by **40–60%**.
- Designed a **3-service microservices architecture** (React, Express, FastAPI) in a **TurboRepo** monorepo, using **PostgreSQL** and **Gemini 2.5 Flash** with automatic local-LLM fallback for zero-downtime evaluation.
- Implemented an **ambient AI code-review** feature using **WebRTC data channels**, automatically analyzing candidate code every **2 minutes** to deliver proactive, context-aware feedback in real time.

Digital Payments Application

Source Code

- Designed monorepo architecture with **Turborepo** for efficient code sharing and service isolation.
- Built a secure server-side **Webhook** system for handling bank API callbacks, preventing race conditions and unauthorized balance updates.
- Built a secure real-time peer-to-peer payment system with **email-verified** user onboarding using time-based JWT authentication, enabling **instant transfers**, payment requests, and **ACID-compliant transactions** ensuring data integrity and consistency.
- Developed user **dashboard** with directory, instant payment actions, real-time **wallet updates**, and transaction management including **history tracking**, monthly filters, and request queue.

Technical Skills

Coding Languages: C, C++, Python, JavaScript, TypeScript

Backend & Serverless Development: Node.JS, Express.JS, Next.JS, Cloudflare Workers, RabbitMQ

Frontend: React.js, Next.JS, TailwindCSS, HTML, CSS

Databases & ORM: PostgreSQL, MongoDB, Prisma ORM

Developer Tools: Postman, GitHub, Docker, TurboRepo

Technical Activities

ACM, SVNIT

2025 – Present

Executive Member

- Actively participated in coding workshops and technical sessions organized by the club
- Gained exposure to problem-solving approaches and basic competitive programming concepts

Coding Achievements

- Solved over 700+ DSA problems across various platforms including LeetCode, Codeforces, and GFG.
- Pupil** on Codeforces.(1310)
- LeetCode(1577)